STORAGE DEVICES

Your computer’s memory box

Bryan L. Tarazona R.



So, what does storage do?

Storage keeps your files even when the power is off. Photos, games, apps and the operating system all live there.



Meet the three main drives

01

HDD

Spinning disks

Cheap + roomy



02

SATA SSD

Flash memory

Fast upgrade



03

NVMe SSD

Flash + PCIe

Fastest choice



Also around: USB drives • memory cards • cloud storage

HDD: the spinning disk

Inside, shiny platters spin. A tiny arm reads and writes data — a bit like a record player.

✓ Lots of space for less money
× Slower, noisier, easier to damage



SSD: no moving parts

SATA SSD

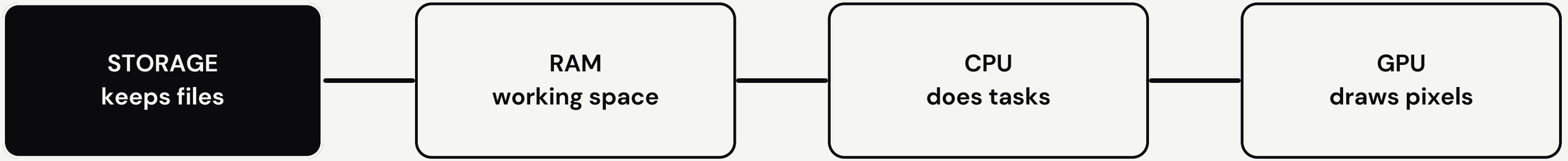
Uses a cable
Up to ~550 MB/s
Great older-PC upgrade

NVMe SSD

Plugs into the motherboard
Up to ~7,000 MB/s
Tiny and super fast

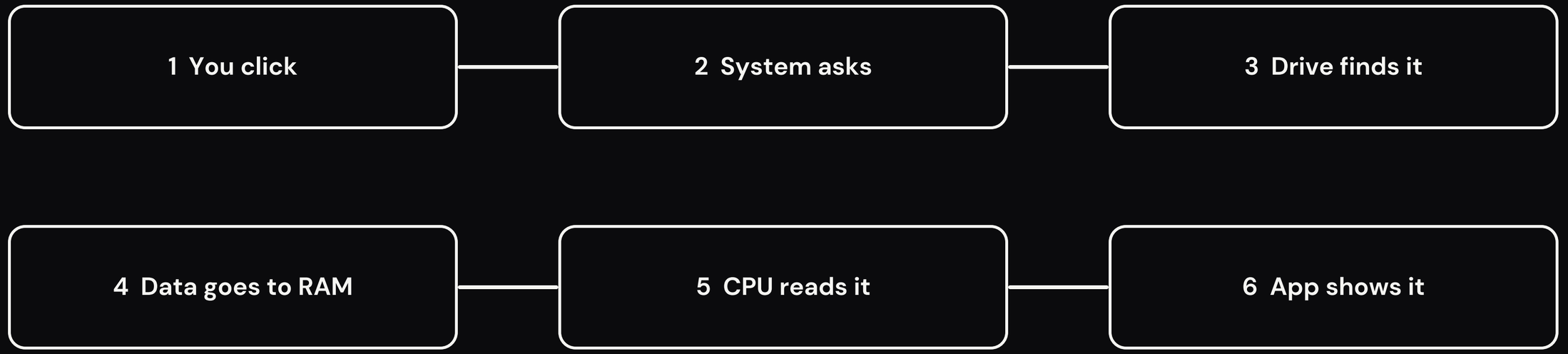
Same memory idea. Different road to the CPU.

Teamwork inside the computer



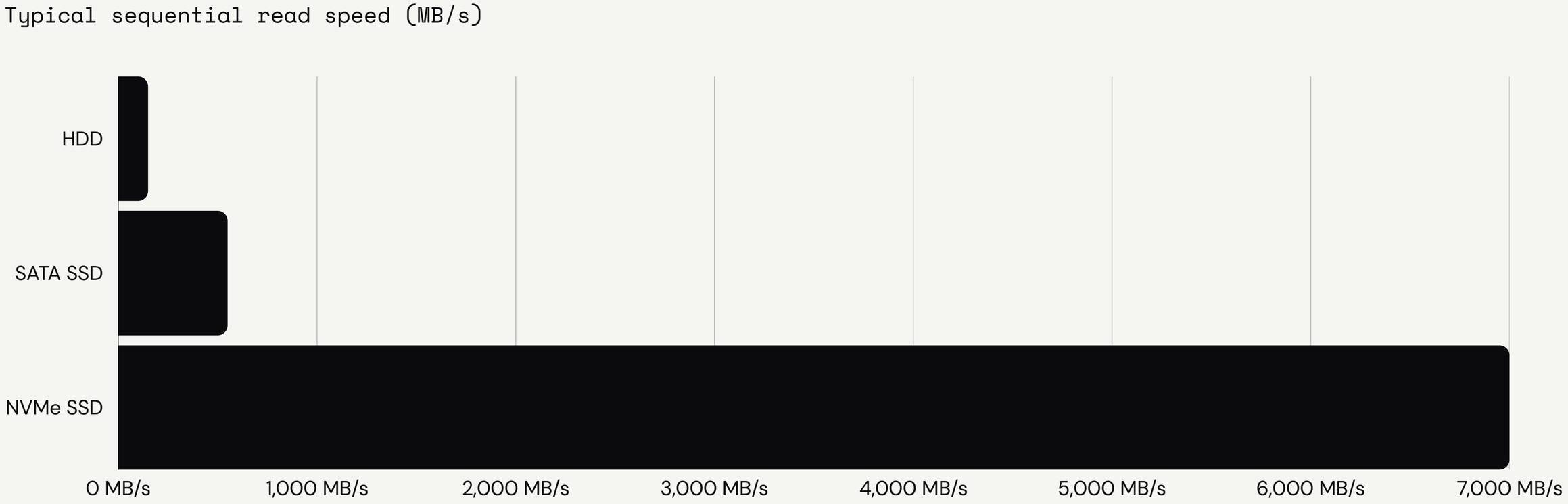
Storage keeps it. RAM holds the current work. The CPU does the thinking. The GPU draws the picture.

What happens when you open a file?



Fast drive = less waiting.

Who wins? The numbers



HDD • cheapest • 1–8 TB

SATA SSD • balanced • 500 GB–4 TB

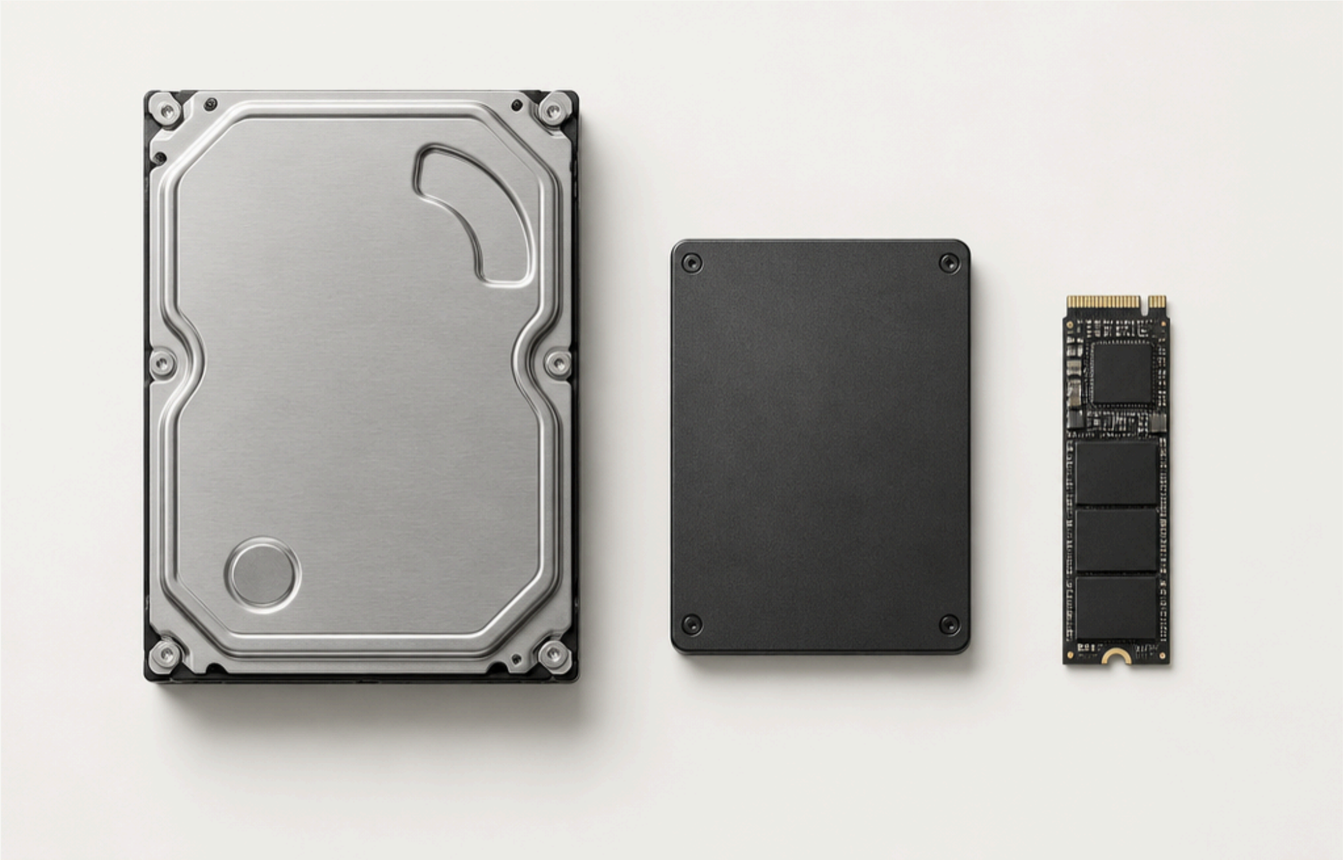
NVMe SSD • fastest • 500 GB–4 TB

Shapes and sizes

3.5-inch
Desktop HDD

2.5-inch
Laptop HDD or SATA SSD

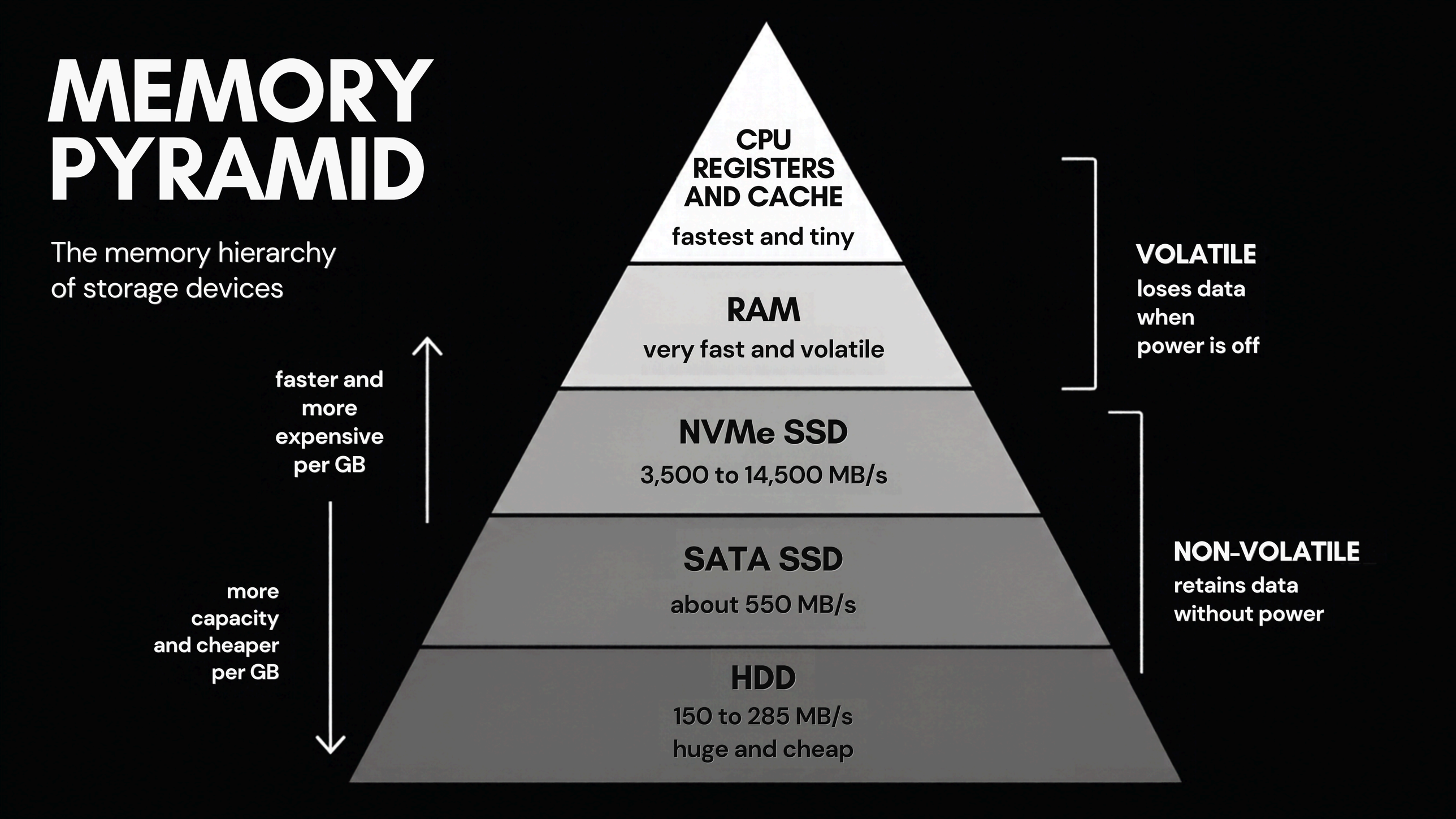
M.2
Slim NVMe ‘stick’



Always check the slot before you buy.

MEMORY PYRAMID

The memory hierarchy of storage devices



Real drives you can buy

BarraCuda 3.5" 2TB SATA 3 Seagate ≈ €149.95

SATA SSD 870 EVO 1 TB Samsung ≈ €294.99

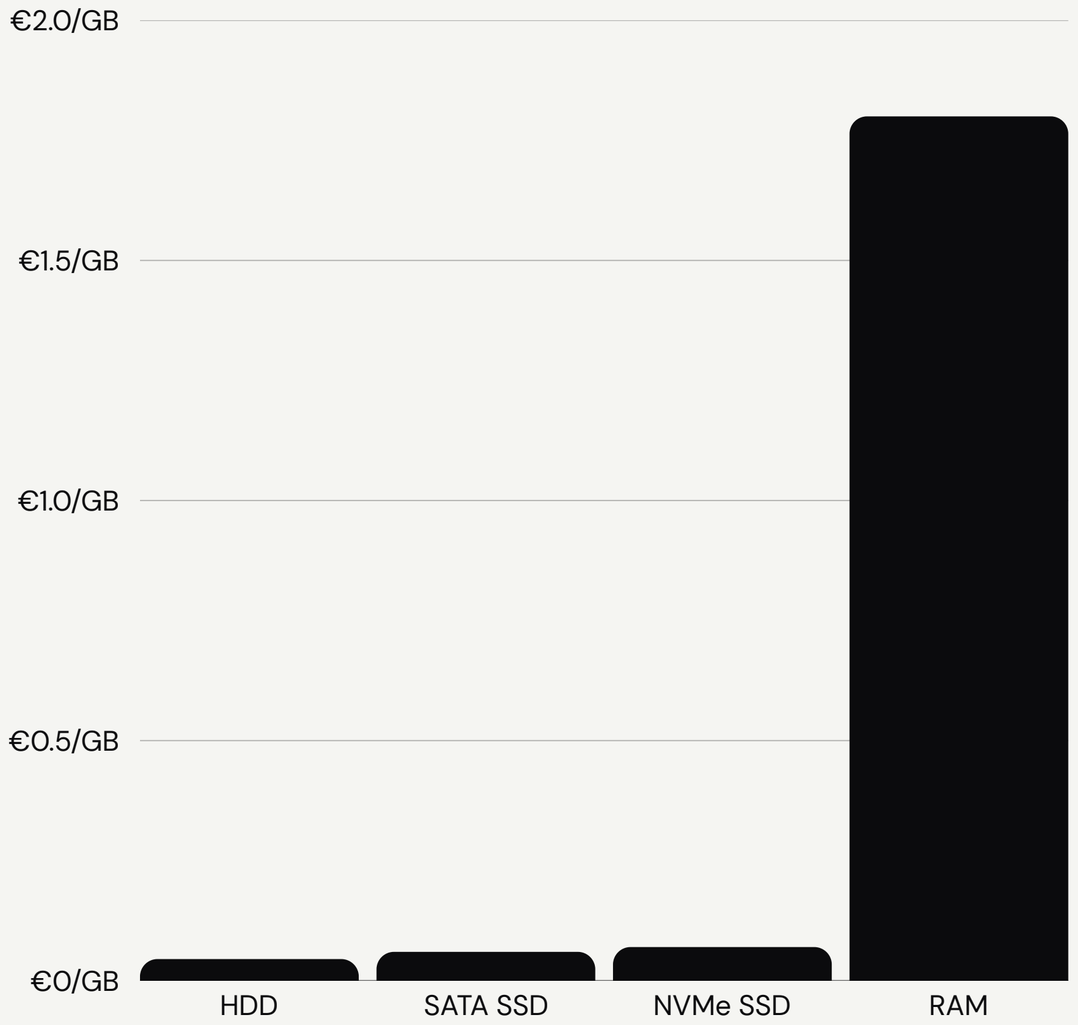
NVMe SSD 990 EVO Plus 1 TB Samsung ≈ €229.56



PcComponentes Spain, Price checked on 29 october 2026

Fun fact: storage vs RAM price

Storage keeps data long-term. RAM is much faster — and much more expensive per GB.



RAM can cost 40× more per GB than an HDD.

QUICK RECAP

HDD = big + budget

SATA SSD = easy + quick

NVMe SSD = tiny + fastest

Which drive would you choose — and why?

TIME TO KAHOOT